

PRACHI MITTAL

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EDUCATION

Master of Science, Information Technology

Arizona State University, Tempe, AZ

May 2025

GPA: 3.83/4.00

Master of Computer Applications

Graphic Era University, India

May 2021

CGPA: 9.58/10.00

TECHNICAL SKILLS

Programming: Python, Java, C#, SQL

Core CS: Data Structures, Algorithms, Distributed Systems

Backend: REST APIs, Microservices, .NET Core, Flask

Cloud: AWS (Lambda, EC2, S3, SQS, SNS, API Gateway), Docker, Kubernetes

Databases: PostgreSQL, MySQL, DynamoDB

CI/CD: Jenkins, GitHub Actions, Azure DevOps

AI/ML: LLM Integration, RAG, Vector Search, NLP

PROFESSIONAL EXPERIENCE

Cloud Engineer — American Tire Distributors

Nov 2025 – Present

Charlotte, NC

- Designed and built a Kubernetes-based microservice to process documents through asynchronous workflows, implementing concurrency control and retry mechanisms for scalable and fault-tolerant execution, reducing processing time by over 95% (45 min → <1 min).
- Developed backend services using Python and PostgreSQL to replace a \$230K/year payroll system, improving data control and eliminating external dependencies.
- Developed REST APIs and data pipelines, identifying and resolving race conditions in concurrent request handling, improving system reliability and increasing throughput by 30%.
- Automated data processing workflows, reducing manual intervention by 40% and improving end-to-end processing efficiency.

AWS Cloud Developer — MindSpark

Jun 2025 – Nov 2025

Tempe, AZ

- Designed and implemented serverless data processing pipelines using AWS Lambda and Step Functions to orchestrate large-scale, fault-tolerant workflows, processing 10K+ events daily with high reliability.
- Developed distributed backend logic to manage data transformations across asynchronous services, improving data processing efficiency and reducing failure rates by 25%.
- Built CI/CD pipelines to automate deployment, reducing release time by 30% and improving deployment consistency for backend systems.

Teaching Assistant — Arizona State University

Aug 2024 – May 2025

Tempe, AZ

- Mentored 250+ students in debugging and problem-solving, focusing on structured approaches to building and troubleshooting software systems.

Software Engineer — Cognizant Technology Solutions

Apr 2021 – Aug 2023

Pune, India

- Developed backend microservices using C#/.NET to ingest and process client and product data from multiple sources, supporting high-concurrency workloads and handling 50K+ client-product evaluations daily.
- Designed and implemented a rule-based decision engine to evaluate client eligibility for financial products (mutual funds, ETFs) based on complex cross-border regulatory constraints, improving decision accuracy and reducing manual processing by 40%.
- Built data mapping pipelines to align client and product attributes for scalable matching.
- Optimized data processing workflows, improving throughput and reducing latency in high-volume systems.

PROJECTS

AutoAssist AI — Intelligent Chatbot (RAG System)

- Designed and built a retrieval-augmented chatbot integrating vector search and LLMs for real-time query resolution over dynamic datasets.
- Developed backend APIs and data pipelines to support scalable ingestion and retrieval of structured and unstructured data.

FeedbackIQ — Cloud-Native Feedback Platform

- Designed an event-driven serverless architecture to process 1000+ daily feedback inputs using asynchronous pipelines.
- Implemented real-time sentiment processing and scalable data workflows, reducing manual effort by 60%.